EN4554 Assignment 2 on CNNs

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Setup

Write your own Jupyter notebook (PyTorch). It must run on a Colab T4; on a laptop. Note that running all the tasks will take a few hours. Use CIFAR-10 (`torchvision.datasets.CIFAR10`): split the training set into 45k train and 5k validation, normalize per channel with training statistics, and fix random seeds. Use the test set only once, in Task 4. In every task, interpret your plots and tables in a few sentences; the interpretation carries marks.

Task 1: Backpropagation

(a) Implement Linear, ReLU, and softmax cross-entropy with explicit `forward`/`backward` (no autograd), and assemble an MLP.
(b) Verify your gradients against autograd and against centred finite differences; report the relative errors. Use torch.allclose to check whether the gradients match.
(c) Train it with SGD on flattened CIFAR-10 images; plot the loss and report validation accuracy.
(d) For a 10-layer MLP, plot the per-layer gradient norm at initialization with sigmoid and with ReLU. Explain.

Task 2: Optimizers

(a) Implement SGD with momentum and Adam (with bias correction) as `torch.optim.Optimizer` subclasses. Show that they match `torch.optim.SGD` and `torch.optim.Adam`.
(b) On $f(x, y) = \frac{1}{2}(x^2 + 50y^2)$, plot the trajectories of SGD, SGD with momentum, and Adam. Use contour to plot the contours of $f(x, y)$ Explain the differences.
(c) Train a small CNN with each optimizer (plain SGD, SGD with momentum, Adam) at three learning rates each. Plot validation accuracy versus learning rate and the training loss at the best rate. Discuss sensitivity.

Task 3: Regularization and residual connections

(a) Train a small CNN with batch norm on 5k training images. Compare: baseline, weight decay, dropout, flip+crop, cutout, mixup, and all combined (implement cutout and mixup yourself). Tabulate train accuracy, validation accuracy, and their gap; plot the curves.
(b) Build CIFAR-style plain and residual networks of depth $6n + 2$ for depths 8, 20, and 56 (three stages, channels 16/32/64, global average pooling, batch norm, Kaiming initialization). The only difference between the two families is the shortcut. Plot the training loss versus epoch and explain.

Task 4: CNN (Optional)

Design the best CNN you can with at most 500k parameters and at most 20 training epochs. Tune on the validation split. Report an ablation of at least three design choices (e.g., shortcut, augmentation, weight decay, optimizer), and the test accuracy of your final model, evaluated once.

GitHub Profile

You must include the link to your GitHub (or some other SVN) profile, so that I can see that you have worked on this assignment over a reasonable duration. Therefore, make commits regularly. However, I will use only the pdf for grading to save time.

Claude or Copilot Use

You may use these tools to help you with coding, but you must fully understand the code. I may include questions about the code in the final examination. You must acknowledge the use of these tools in your report.

Submission

Upload a pdf of your Jupyter Notebook named as your_index_a02.pdf. Include the index number and the name within the pdf as well. The report must include important parts of code, image results, and comparison of results. The interpretation of results and the discussion are important in the report.